Original Article

Evaluation of Adaptation Methods and Cognitive Dysfunction in Multiple Sclerosis Patients

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ABSTRACT

Cognitive impairments in patients are less frequently diagnosed, which itself leads to patients using inappropriate adaptation methods. The aim of this study was to evaluate adaptation methods and cognitive dysfunction in multiple sclerosis (MS) patients. In this descriptive cross-sectional study, 100 patients with MS referred to the Neurology Clinic of Alavi in Ardabil, Iran, were evaluated. The patients completed Lazarus' Coping Strategies Questionnaire, the Adult Mini-Mental State Examination (MMSE), and the Digit Symbol Coding Test, which were used to assess adaptation and coping methods, cognitive disorders, and processing speed, respectively. The mean age of the MS patients was 44.49±10.16 years. 76% of the participants were female, 42% were illiterate, 78% were married, and 69% were employed. The mean disease duration was 13.53±9.9 years. Among the various adaptation methods, "Accepting Responsibility" (17%) and "Seeking Social Support" (15%) were the most frequently reported, while "Positive Reappraisal" (8%) was the least used method by the patients. Routine assessment of cognitive abilities in patients with MS can help prevent their cognitive deterioration and design effective pharmacological and non-pharmacological interventions.

Introduction

Multiple sclerosis (MS) is considered one of the most common neurological disorders of the central nervous system in young adults [1]. In addition to the sensory and motor disorders that manifest in MS, cognitive impairment occurs frequently and affects approximately 40 to 70 percent of MS patients [2]. Furthermore, cognitive impairment can appear at the beginning of the disease and be the first manifestation of MS [3]. Although full-blown dementia rarely occurs in MS patients, even mild cognitive impairments can cause significant functional disability for the patient. MS patients with cognitive impairment have difficulty finding employment, participate in fewer social activities, experience significant hardship in performing daily tasks, benefit less

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from rehabilitation therapies, and have more psychopathology compared to MS patients without cognitive problems [4,5]. During the treatment of these patients, their neuropsychological profile is not fully considered, and cognitive problems are less frequently diagnosed, leading to numerous difficulties for MS patients. This, in turn, leads to the development of adaptation methods in these patients. The aim of this study was to investigate adaptation methods and cognitive problems in MS patients referred to the Alavi Educational-Therapeutic Center in Ardabil.

Methods

This descriptive cross-sectional study was conducted on 100 MS patients referred to the neurology clinic of Alavi Hospital in Ardabil from October 2023 to October 2024. Patients entered the study upon confirmed diagnosis of MS based on a neurologist's opinion and had a medical record in the neurology clinic. Patients with comorbid psychotic disorders, intellectual disability, and mental or motor disabilities were excluded from the study. Upon providing written consent to participate in the study, patients completed two questionnaires: the Lazarus Coping Strategies questionnaire and the Adult MMSE, which assessed adaptation and coping methods and cognitive disorders in MS patients, respectively. Additionally, due to the low sensitivity of the MMSE for MS patients, the Symbol Digit Modalities Test (SDMT) is sensitive, but SDMT test was also used to better evaluate cognitive problems in these patients.

SDMT Test

Processing speed and working memory of patients were assessed using the SDMT test. This test contains 9 symbols paired with numbers 1 to 9. The patient sees the paired symbols with their corresponding numbers at the top of the page. At the bottom, the symbols are arranged randomly. The patient must name the numbers corresponding to the unpaired symbols at the bottom of the page as quickly as possible. The number of correct numbers named in 90 seconds was recorded and evaluated based on normal cut-off values for different age groups.

Lazarus and Folkman Coping Strategies Test

The Lazarus Coping Strategies questionnaire, based on the Ways of Coping Inventory, developed by Lazarus and Folkman in 1980 and revised in 1985, assesses coping strategies, which are a set of cognitive and behavioral efforts an individual uses to interpret and modify a stressful situation, leading to a reduction in the distress caused by it. It evaluates a wide range of thoughts and actions that individuals employ when facing internal or external stressful conditions. The components of the test include confrontive coping, distancing, self-controlling, seeking social support, accepting responsibility, escape-avoidance, planful problem-solving, and positive reappraisal. The scores obtained from the questions for each component determine the score for that specific component, and the component with the highest score was reported as the individual's dominant adaptation strategy. The Lazarus questionnaire has been standardized in Iran on a sample of 750 middle-aged couples and a sample of 763 high school students. In reliability assessment, Cronbach's alpha coefficient for the scales was reported as desirable, ranging from 0.61 to 0.79.

MMSE Test

It consists of three 5-point items, three 3-point items, one 2-point item, and four 1-point items. Its 5-point items include time orientation, place orientation, and serial sevens. Its 3-point items include: following a three-stage command (*e.g.,* stand up, walk, and sit down), recent memory, and immediate recall. Its 2-point item is naming two objects, with each object worth one point. Its 1-point items include reading, writing, repetition, and design. A score of 25-30 was considered normal, 21-24 mild cognitive impairment, 10-20 moderate cognitive impairment, and scores less than 10 severe cognitive impairment. The validity and reliability of this test were confirmed with an overall Cronbach's alpha coefficient of 0.81. A cut-off value of 22 for this questionnaire was associated with a specificity of 93.5% and sensitivity of 90%.

Statistical Data Analysis

Data from the questionnaires, as well as data from the SDMT test and demographic information of the patients, were collected in a checklist and entered into SPSS software version 26. Data were expressed using descriptive statistics methods in the form of tables and graphs. Statistical tests such as t-test and ANOVA were used to examine the differences in cognitive disorders and adaptation methods based on demographic variables. The significance level was considered 0.05 in all cases.

Ethical Considerations

Patient names were not used in the study. The study was conducted after obtaining necessary permissions from the ethics committee (Ethics Code: IR.ARUMS.REC.1399.421).

Results

The mean age of the MS patients was 44.49 with a standard deviation of 10.16 years. 76% of the participants were female, 42% were illiterate, 78% were married, and 69% were employed. The mean age at disease onset was 30.96 years with a standard deviation of 4.27, and the mean disease duration was 13.53 with a standard deviation of 9.90 years (Table 1).

According to the statistical test results, there was no significant difference in the mean SDMT processing speed scores based on demographic variables (Table 2).

The mean of MMSE test score in males was significantly lower than females (p=0.023), but there was no significant difference in mean MMSE test results among MS patients based on other demographic variables (Table 3).

Table 1: Frequency of demographic variables

Variable	Category	n	%
Age group	Under 30 years	8	8
	30–39 years	34	34
	40–49 years	29	29
	50–59 years	21	21
	60 years and above	8	8
Gender	Male	24	24
	Female	76	76
Education level	Illiterate	42	42
	Primary school	23	23
	Secondary and high school	20	20
	University	15	15
Marital status	Single	22	22
	Married	78	78
Employment status	Employed	69	69

Variable	Category	n	%
Age at onset of disease	Unemployed	31	31
	Under 30 years	42	42
	30–39 years	58	58
Duration of disease	<5 years	25	25
	5–15 years	39	39
	15–25 years	23	23
	>25 years	13	13

Table 2: SDMT test results based on demographic variables in patients with MS

SDMT results score variable		Mean	Standard deviation	*P*-value
Age group	<30 years	52.13	17.58	0.145
	30–39 years	55.71	18.84	
	40–49 years	52.48	14.65	
	50–59 years	59.48	13.70	
	60+ years	52.38	12.65	
Gender	Male	60.00	14.86	0.08
	Female	53.43	16.18	
Employment	Employed	56.28	16.45	0.754
	Unemployed	52.19	15.01	
Age at onset	<30 years	58.33	16.63	0.366
	30–40 years	52.60	15.32	
Disease duration	<5 years	51.08	17.40	0.544
	5–15 years	56.13	17.78	
	15–25 years	55.74	12.21	
	>25 years	57.92	14.10	
Education level	Illiterate	59.57	14.03	0.731
	Primary	50.57	14.95	

SDMT results score variable		Mean	Standard deviation	*P*-value
Marital status	Secondary/high school	54.45	20.09	0.599
	University	49.80	14.90	
	Single	53.77	15.52	
	Married	55.36	16.28	

Table 3: MMSE test scores based on demographic variables in MS patients

MMSE score results variable		Mean	Standard deviation	*P*-value
Age group	<30 years	23.38	4.07	0.141
	30–39 years	24.85	2.90	
	40–49 years	25.07	2.81	
	50–59 years	25.00	4.21	
	60+ years	26.38	2.97	
Gender	Male	23.63	3.97	0.023*
	Female	25.37	2.95	
Employment	Employed	24.84	3.37	0.088
	Unemployed	25.19	3.13	
Age at onset	<30 years	24.31	4.02	0.128
	30–40 years	25.41	2.57	
Disease duration	<5 years	25.04	2.78	0.578
	5–15 years	24.67	3.27	
	15–25 years	25.30	2.87	
	>25 years	25.00	4.92	
Education level	Illiterate	24.74	2.91	0.727
	Primary	25.09	3.25	
	Secondary/high school	24.70	4.31	
	University	25.67	2.99	

MMSE score results variable		Mean	Standard deviation	*P*-value
Marital status	Single	24.73	3.03	0.481
	Married	25.01	3.37	

**p<0.05;t-test.*

The most common type of coping strategy among patients with MS was acceptance of responsibility (17%), followed by seeking social support (15%) (Figure 1).

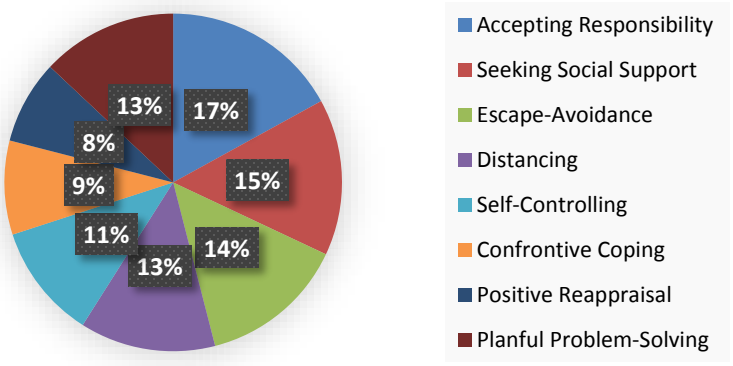


Figure 1: MS patients participating in this study classified by type of coping strategy with the disease

In men, according to the Lazarus coping strategies test, the most prevalent strategies were escape-avoidance and positive reappraisal, each observed in 5 cases (20.8%).
In women, the most frequent coping strategy was acceptance of responsibility (14 cases, 18.4%), followed by seeking social support and self-control, each reported in 11 cases (14.4%). There was a statistically significant difference between men and women regarding the type of coping strategy used (p = 0.041) (Figure 2).

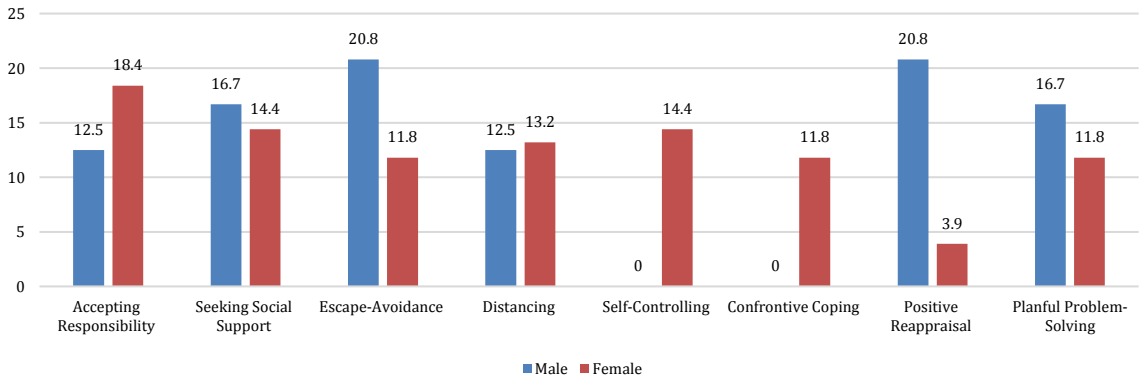


Figure 2: MS patients participating in this study classified by type of coping strategy based on gender

The mean score of participants in the SDMT (Symbol Digit Modalities Test) was 47.14 ± 10.17. According to test interpretation, 10% of the participants were below the expected level, and the rest were within the normal range (Figure 3).
The mean score obtained from the MMSE test was 24.95 ± 3.29 (Figure 4).

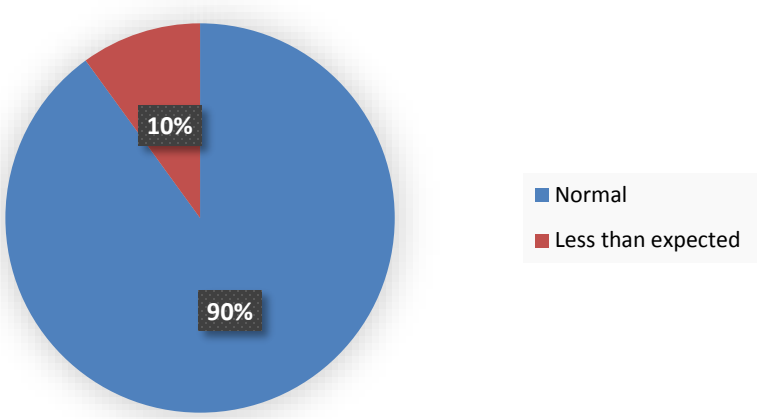


Figure 3: Results of the SDMT test in MS patients

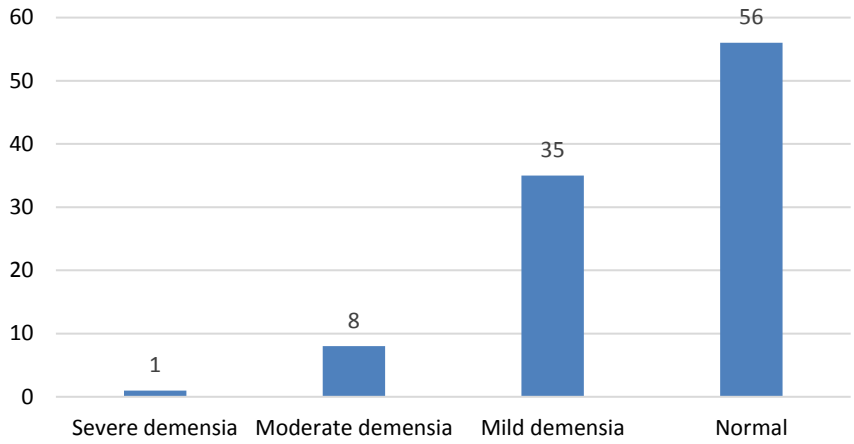


Figure 4: Results of the MMSE test in MS patients

Discussion

According to the study results, the mean age of participants was 44 years, with 76% being female and 24% male. The mean age at disease onset was 30 years, and the mean disease duration was 13 years. Based on the Lazarus coping strategies test results, the distribution of coping methods was as follows: accepting responsibility in 17%, seeking social support in 15%, escape-avoidance in 14%, distancing and planful problem-solving each in 13%, self-controlling in 11%, confrontive coping in 9%, and positive reappraisal in 8% of patients. According to the results, the highest rate of cognitive impairment was reported in females and in the 30-40 age range.

Significant psychiatric complaints and neurobehavioral symptoms of this disease include depression, forgetfulness, mood disorders, and cognitive disorders, such as impaired cognition and poor memory [7]. Studies have shown that depression is the most common disorder in MS patients, often being a reaction to the disease course. Furthermore, medications used for treatment are factors in causing or exacerbating depression. Depression also severely impacts the reduction of quality of life [8].

In recent decades, various studies have been conducted on stress management and its effects on promoting mental and physical health. The results of these studies indicate the direct and indirect impact of stress management on improving psychological disorders and physical problems. In fact, this treatment program was primarily created for individuals grappling with stress resulting from severe chronic physical illnesses. This intervention comprises elements such as increasing awareness about stress, relaxation training, identifying inefficient thoughts, cognitive restructuring, problem-

solving training, and assertiveness skills training. Stress is a factor that has a bidirectional relationship with MS. Given that MS is a somewhat unknown and unpredictable disease, it increases stress and anxiety in patients; on the other hand, stress also exacerbates the symptoms of MS [9].

In the experience of many individuals with MS, there is a strong association between stress and disease relapse, and most patients report worsening symptoms during times of stress. Over time, stress requires more energy for thinking and solving daily life problems, and this energy depletion leads to fatigue and impaired daily functioning in patients [10]. Disease exacerbation is unpredictable, and in 85-90% of cases, periods of symptom exacerbation are observed. It is now known that psychological stressors activate this disease [11]. One important psychological dimension in MS patients, affected by the disease and the stresses arising from the perception of the illness, is their level of resilience. Studies show that not only is the resilience of MS patients lower than that of healthy individuals, but it is also lower than that of patients with other chronic diseases [12]. Resilience can foster hope in clinical experiences, and its major advantage is that it enables individuals not to easily surrender in the face of difficult situations [13]. Resilience is one of the most important human abilities that leads to effective adaptation to risk factors and, in medicine and psychology, indicates physical resistance, spontaneous recovery, and the ability to re-establish emotional balance in stressful situations. It essentially refers to the characteristics that affect the speed and extent of recovery after encountering stress. Studies have shown that individuals with high resilience exhibit lower levels of avoidance and cope with the pain caused by their disease. They do not have a catastrophic attitude toward pain and are less inclined toward catastrophizing [14]. Overall, resilience involves maintaining composure under pressure, flexibility in facing obstacles, avoiding debilitating strategies, maintaining optimism and positive emotions during hardships, and overcoming internal barriers. It can moderate stress levels and disability in stressful conditions and increase life satisfaction [15]. Resilience does not eliminate stress or remove life's problems, but it empowers individuals to have more adaptive coping styles in the face of problems. Thus, resilience, as a dynamic psychological process, influences individuals' responses in various life situations, especially critical times, and by controlling emotions and feelings, leads to better performance in these situations [16].

Since MS patients must cope with both daily life stresses and the stresses arising from the fluctuating and unpredictable symptoms of the disease, disease progression may interfere with work, family life, relationships, and social activities. The goal of psychological interventions is to help these individuals cope with the aforementioned challenges. Norte *et al.*, in a study, investigated the effectiveness of cognitive-behavioral stress management intervention on resilience, anxiety, and depression in patients with post-traumatic stress disorder. The results showed that cognitive-behavioral stress management interventions were effective in increasing resilience and reducing anxiety and depression [17].

Emami *et al.*, in a study, examined the effectiveness of group cognitive-behavioral stress management therapy on the resilience and hope of women with breast cancer. The results indicated that cognitive-behavioral stress management intervention led to increased resilience and hope in the experimental group of women with cancer, and a one-month follow-up confirmed the stability of the treatment [18].

Tabibzadeh *et al.*, in a study, investigated the effectiveness of cognitive-behavioral therapy on the resilience and rumination of MS patients. The results of the analysis of covariance showed a significant difference between the experimental and control groups in the level of resilience and reduction of rumination after the intervention, and these effects remained stable after one month [19].

Chermahini *et al.* also examined the effectiveness of group cognitive-behavioral therapy on the anxiety and resilience of menopausal women. The results showed that cognitive-behavioral therapy reduced anxiety and increased resilience in menopausal women [20].

To explain the above findings, it can be mentioned that the reason for the effectiveness of cognitive-behavioral intervention on increasing adaptation seems to be, from a functional perspective, learning how to face inevitable life problems and coping better with stressful conditions (including chronic illness) and unpleasant events [21].

Furthermore, by emphasizing cognitive components, including negative attitudes and thoughts, special focus is placed on replacing unrealistic cognitive appraisals with realistic ones. This approach can help free individuals from unwanted thoughts, habits, and unhealthy behavioral patterns. Consequently, the patient becomes aware of their abilities, and their adaptation, through role exchange in life, increases. By increasing cognitive-behavioral skills, the individual develops a stronger sense of control. This sense convinces individuals to take charge of uncontrollable situations. This intervention, by focusing on the components of commitment, control, and challenge, increases hardiness and consequently enhances adaptability. In fact, by creating realistic appraisals and appropriate cognition, it causes the individual to employ desirable coping methods to remove or control the stressor and, by changing their cognition about themselves and their abilities, to let go of self-alienation and have a sense of self-worth. This ultimately increases the individual's sense of commitment. Additionally, by teaching desirable coping strategies and replacing problem-focused strategies instead of emotion-focused ones in patients, it strengthens their sense of challenge; so that individuals consider life's problems as a field for struggle and growth and do not show a state of helplessness and surrender in the face of events. If stress is managed and effective coping skills are provided, the individual will be able to better cope with the needs and challenges of their life and exert more commitment, challenge, and control in their life [22]. The stress management program combines various types of relaxation, imagery, and other anxiety reduction techniques with common cognitive-behavioral approaches, such as cognitive restructuring, coping effectiveness training, assertiveness training, and anger management. Using these techniques can reduce stress levels and help promote health and improve immune and social functioning in patients [23]. Furthermore, controlling and reducing stress using relaxation methods and cognitive techniques leads to increased adaptation in patients because learning meditation, relaxation, and cognitive methods enables the individual to face stress and life problems effectively and efficiently. Another factor is conducting the therapy in a group format. Many experts believe that group therapy is very effective. The group improves the patient's communication skills and provides them with a useful support system. On the other hand, the patient, by observing others, realizes that their problem is not unique [24]. Limitations included the reliance on self-report tools to determine stress adaptation methods and the concurrent use of chemical medications by participants, which could have influenced the treatment interventions.

Conclusion

Based on the obtained results, the routine assessment of cognitive abilities in MS patients can help prevent their cognitive deterioration and design effective pharmacological and non-pharmacological interventions. On the other hand, training in effective adaptation and problem-solving methods can reduce their stress levels and improve their treatment process. It is suggested that similar studies with the following characteristics be conducted based on this review, Given the high prevalence of MS in females, it is recommended that a similar study with a larger sample size be conducted specifically on females. Considering the role of MS treatment medications in causing cognitive and psychiatric disorders, it is recommended that the impact of drug therapy on the development of these changes be investigated. Conducting a similar study on a larger sample size, as well as investigating the prevalence of psychiatric disorders in patients with MS and simultaneously examining the effect of drug intervention and cognitive-behavioral interventions on improving the quality of life of those with MS.

Conflict of Interest

The authors declared that there was no conflict of interest in this work.

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